

Sweep Gas Flow Titration

Adjustments to sweep gas flow are probably the most common adjustments that you will make in the course of an extracorporeal membrane oxygenation (ECMO) run. While it is entirely feasible that you can go for days on the same blood flow, sweep may need to be adjusted on a daily basis or even multiple times per day.

There is little evidence on optimal strategies for sweep titration and there is significant variation in practice patterns. That said, having a structured approach to adjusting sweep can shape the course of the ECMO run and can optimize management of patients on ECMO. As such, the strategies presented should not be taken as definitive but rather as a proposed approach to the physiologic principles at play.

In general, sweep gas titration is a skill set that gets better with practice. Every adjustment that is made to sweep parameters is an opportunity to assess the clinical impact. As with all critical care interventions, you should have a rationale for changes, estimate the expected result, and then reassess the clinical response. This approach will hone your intuition and serve to refine your clinical thought processes. Remember, practice doesn't make perfect, perfect practice makes perfect.

BASICS OF SWEEP GAS FLOW TITRATION

Recall that the oxygenator membrane is semi-permeable, allowing for O_2 and CO_2 to diffuse down their respective concentration gradients and that while both molecules are able to move across the membrane, the diffusibility of CO_2 is six times greater than the diffusibility of O_2 (Fig. 18.1).

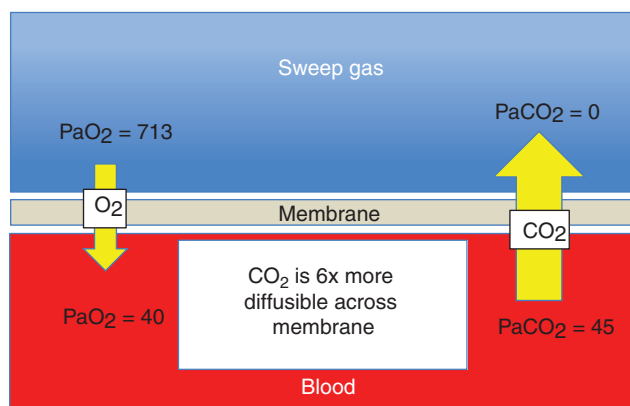


FIG. 18.1 Relative diffusibility of CO_2 v. O_2 across the ECMO membrane

This higher diffusibility of CO_2 means that the rate of gas flow directly correlates with how fast CO_2 is cleared. The higher the rate of gas flow, the faster CO_2 is removed from the blood.

This concept underlies the convention provided that PaCO_2 is primarily affected by adjustments to **flow rate of sweep gas**, while PaO_2 of the blood leaving the oxygenator is primarily affected by adjustments to the FiO_2 of the sweep gas flow (Fig. 18.2).



FIG. 18.2 Manipulate sweep gas flow rate to control PaCO_2 and FiO_2 to control PaO_2 . (From PIJITRA PHOMKHAM/Shutterstock.com.)

For the most part, adjustments of 1 L or greater should be made with the middle knob (increments of 0.5 L) while adjustments to sweep of <1 L should be made with the knob on the left (increments of 100mL), to allow for more precise titration. These two knobs will be additive, so that if the middle knob is set to 4 L/min and the left knob is set to 500 mL/min, the patient will be receiving 4.5 L/min of sweep gas flow.

How much does an adjustment in sweep gas flow affect PaCO_2 of arterial blood? The answer lies in the efficiency of the membrane oxygenator, the presence of recirculation, minute ventilation

As a general rule of thumb, an increase in sweep gas flow by 1 L/min can be expected to decrease PaCO_2 by 5–10 mmHg.

and dead space of the native lungs, and the CO_2 gradient, which is a function of PaCO_2 entering into the membrane oxygenator via the venous blood.

The presence of the aforementioned mitigating factors requires that adjustments to sweep gas flow are put into the correct clinical context. Whenever possible, confirm the result of changes in sweep by monitoring the clinical response of the patient (dyspnea, tachypnea, apnea, encephalopathy, etc.) as well as the patient arterial blood gas when clinically appropriate.

PLACING SWEEP GAS FLOW INTO THE CLINICAL PHASE OF ECMO SUPPORT

“Easy enough, when CO_2 is high, go up on sweep and when CO_2 is low go down on sweep, right?” While this is sometimes the case, there is often more subtlety to the titration of sweep gas flow.

The first consideration is what phase of ECMO support the patient is in. While the clinical characteristics of an ECMO run can vary, with setbacks and improvements throughout, we can expect an overall pattern involving three distinct phases: stabilization, maintenance, and weaning. The way we adjust and titrate sweep gas flow will differ significantly depending on what phase we are in. Let's use the graph in Fig. 18.3 to illustrate.

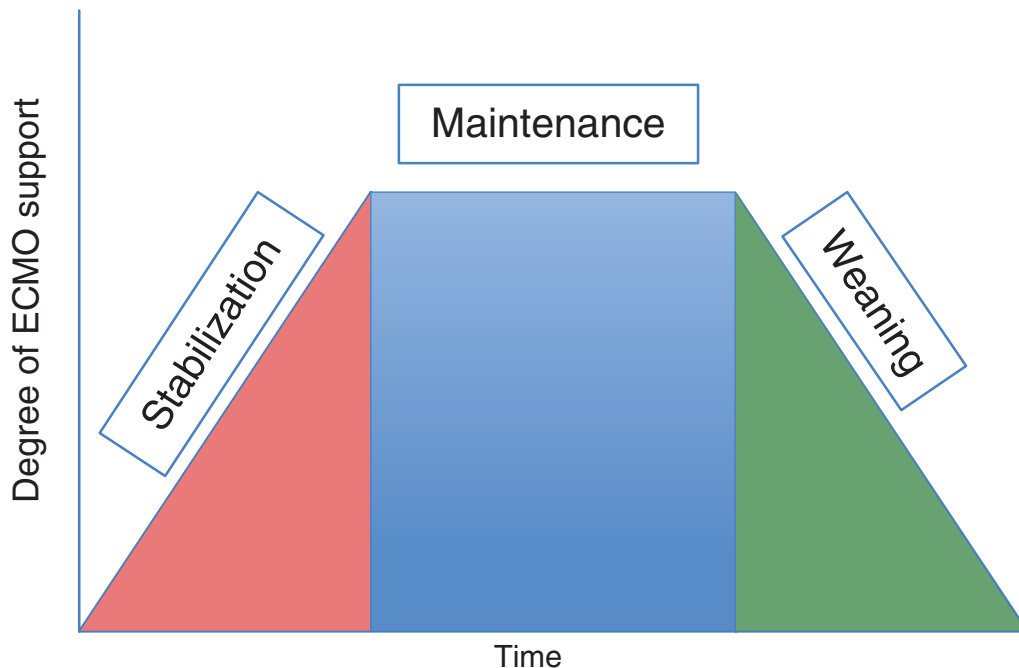


FIG. 18.3 Placing sweep gas flow adjustments into the context of the phase of ECMO support

Stabilization: Stabilization is the period where we are trying to manage and mitigate end organ damage. We are trying to prevent the vicious cycle of cardiac and respiratory failure that could lead to multiorgan failure, with progressive and sequential shutdown of peripheral tissue beds.

During this period, sweep gas flow titration tends to be much more aggressive, with an emphasis on correcting acidosis as well as CO_2 in order to optimize right ventricular afterload, cerebral perfusion, as well as perfusion to end organ tissue beds.

When initiating ECMO, the common convention is to initiate sweep at a 1:1 rate with blood flow and adjust thereafter based on clinical/laboratory parameters. As an example, if you initiate ECMO and the blood flow is set to 3 L/min, sweep gas can be set to 3 L/min with the plan to check arterial blood gases 20–30 minutes after initiating ECMO.

Maintenance: This is the period where stabilization has occurred but the patient still needs ECMO support. The focus of the overall management shifts from mitigating further decompensation to advancing rehabilitation, optimizing nutrition, weaning sedation, and preventing and managing complications.

The length of this period may range from days to weeks depending on the level of support and the overall course of disease.

Sweep gas flow targets in this phase are aimed towards maintaining lung-protective/ultra-lung-protective ventilation, preventing encephalopathy, and patient comfort by titrating to prevent dyspnea/apnea.

When it comes to sweep gas in this phase, there is a rationale/advantage for parsimony. If the patient is not clinically impacted, uncomfortable, or encephalopathic, and if lung-protective ventilation is maintained, maintaining less sweep allows for the ability to increase when needed and allows for engagement of the lungs in CO₂ clearance. Similar to our discussion with blood flow titration, just because you *can* flow at a higher sweep does not mean you *should* flow higher.

Weaning: This is often the most gratifying phase. As the heart/lungs improve, dead space decreases, and V/Q mismatch improves. Additionally, fluid is mobilized from the pulmonary interstitium into the intravascular space, with improved pulmonary compliance, better minute ventilation, and more CO₂ clearance from the native lungs.

This is the phase where sweep can be weaned more aggressively, allowing the lungs to take on more of the load of CO₂ clearance. Sweep can be downtitrated if patient is in the weaning phase, even if the PaCO₂ is normal or even modestly elevated.

Confirm that you are truly in the weaning phase by assessing as many clinical variables as you can. Some questions to ask include:

- Is the compliance on the ventilator improving?
- Is the chest X-ray clear/improved?
- Is there evidence of heart/lung recovery in terms of oxygenation (improved PaO₂)?
- Is there evidence of mobilization of fluid with increased urine output and net negative fluid balance despite minimal or stable diuretics?
- Has blood flow been decreased with minimal impact on clinical status?

As more and more of these scenarios become true, the assumption that the heart/lungs are recovering becomes an increasingly reasonable proposition.

However, note that this can only be done if the patient is truly improved and the lungs are ready to take on the burden of increased CO₂ clearance. If not, this can lead to injury to the lungs, especially if weaning requires a significant escalation in ventilator settings. Patience is paramount – the heart/lungs will take over when they are ready.

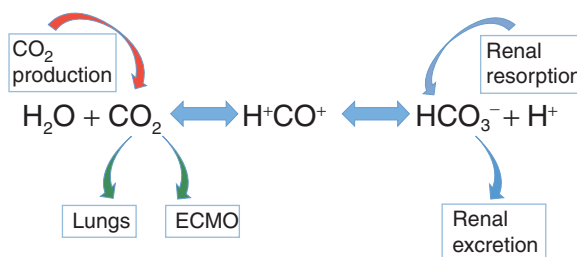
IMPACT OF SWEEP GAS FLOW ON CO₂ BUFFERING

CO₂ dissolves in water to form carbonic acid, which dissociates to bicarbonate/hydrogen ion based on the following equilibrium:



This equilibrium means that excess CO₂ will lead to acidosis in the acute phase, but this acid can be buffered out over time as kidneys resorb HCO₃⁻. This process allows for two phenomena, which are beneficial to the patient with difficulty with CO₂ clearance:

1. Mitigates acidosis from excessive CO₂ through resorption of bicarbonate
2. Allows for two mechanisms to remove CO₂ from the body: lungs (CO₂) and kidneys (HCO₃⁻)



While resorption of excess bicarbonate can take years to accumulate in the case of chronic respiratory acidosis due to chronic obstructive pulmonary disease (COPD) or obstructive sleep apnea, bicarbonate resorption can occur over the course of days in the ICU, particularly in the case of respiratory acidosis due to severe heart/lung failure requiring ECMO.

Therefore, the effect of bicarbonate buffering should be taken into account while adjusting sweep gas flow CO_2 . Let's examine how this manifests clinically.

Adjusting Sweep for pH Versus PaCO_2

If bicarbonate is high due to elevated CO_2 levels, normalizing the PaCO_2 can have the adverse effect of inducing a metabolic alkalosis. This may equilibrate eventually, but there can be adverse effects of changing the CO_2 levels before the patient can adjust. By adjusting sweep gas flow based on the pH rather than the PaCO_2 , you will allow for adjustments in PaCO_2 that are within the set point for the patient.

This convention is typically employed for the maintenance phase. In the resuscitation phase, when there is the possibility of end organ decompensation, normal CO_2 may be targeted in order to prevent cerebral vasodilation/vasoconstriction or to minimize pulmonary vascular resistance/right ventricular afterload.

Adjusting Sweep When pH Is Normal and PaCO_2 /Bicarbonate Is High

Let's consider the case of a patient with worsening CO_2 clearance from the lungs who may also have increasing CO_2 production due to infection, sepsis, or general critical illness. This patient will have elevated total body CO_2 content, with elevated CO_2 as well as HCO_3^- levels, and a consequent normal pH. Increasing sweep alone may lead to metabolic alkalosis. One strategy involves lowering HCO_3^- levels by allowing for dumping of HCO_3^- with a drug like acetazolamide and increasing sweep gas flow if the patient becomes acidotic. The utility of this strategy will depend on the renal function of the patient and an assessment on whether a potential rise in CO_2 will be well tolerated clinically.

Either way, total body CO_2 as a function of PaCO_2 and HCO_3^- needs to be accounted for when making adjustments to sweep gas flow.

SWEEP GAS TITRATION AT THE BEDSIDE

You may be getting a sense of the complexity related to adjustments in sweep gas flow. You may adjust sweep gas very differently for a patient placed on VA ECMO due to cardiac arrest 24 hours ago who is at risk for cerebral edema than a patient on VV ECMO for viral pneumonia who is 40 days into his ECMO run and with a clearing CXR. Whenever possible, place decisions on sweep titration into the context of the total CO_2 burden, to include PaCO_2 and HCO_3^- in addition to the clinical phase of illness. Below is a summary of factors to consider based on the phase of support.

Phase of ECMO support	Reason for sweep increase	Reason for sweep decrease
Stabilization	• Normalize pH and PaCO_2 (optimizing right ventricle and cerebral circulation) • Avoid tachypnea/ dyspnea to whatever extent possible	• Avoid hypocarbia • Alkalotic with normal/ low PaCO_2
Maintenance	• Rest lungs/minimize ventilator settings • Respiratory acidosis • Adverse effects of CO_2: encephalopathy, tachypnea • Worsening oxygenator performance • Normalize bicarbonate	• Respiratory alkalosis • Prevent apnea • Engage lungs in ventilation
Weaning	• Fails weaning trial • Weaned sweep too fast	• Alkalosis or normal pH • Challenge lungs

SWEEP GAS TITRATION IN VA ECMO

The principles outlined earlier apply to VV and VA ECMO, as they dictate the amount of CO_2 clearance that is facilitated by the oxygenator and how this will be translated into the PaCO_2 of blood returning to the body.

Clearly, the way this blood interacts with the body differs between VV ECMO and VA ECMO and should be considered when making adjustments.

VA ECMO allows for the blood to flow directly into the arterial circulation. The effect of this convention is that sweep gas flow is generally weaned much more quickly on VA ECMO support. It is not uncommon to rapidly wean down sweep gas flow to 1 L/min or 0.5 L/min for a patient on VA ECMO.

In fact, patients on minimal sweep (say 0.5 L/min) may continue to have a respiratory alkalosis with a low PaCO_2 and elevated pH. The temptation may be to wean sweep gas flow to off. However, **do not wean sweep gas flow off on VA ECMO!** Doing so will take deoxygenated blood from the venous circulation and dump it into the arterial circulation resulting in a rapid decompensation.

Weaning is generally done differently in VV versus VA ECMO. Let's examine the primary strategies for each.

WEANING IN VA ECMO

Recall that VA ECMO provides direct support for cardiac output in an additive fashion, where overall perfusion provided is the sum of native cardiac output and ECMO blood flow.

Assessment of readiness to wean from VA ECMO involves the ability to wean blood flow. Blood flow can be weaned at bedside in a patient with complete recovery or over the course of days in patients with a more uncertain recovery. The risks of slower wean (longer period at low flows, higher risk of thrombosis) should be weighed against the benefits (more gradual weaning, higher certainty of ability to successfully wean) (Fig. 18.4).

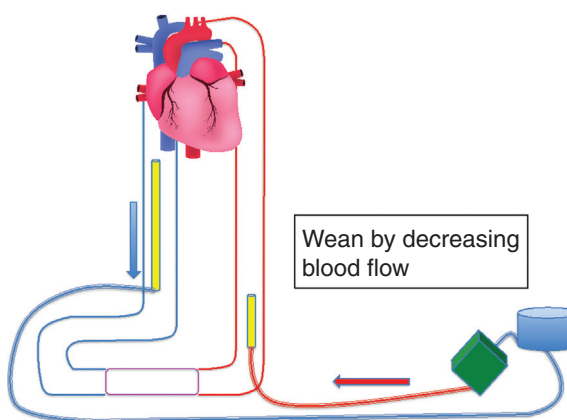


FIG. 18.4 Weaning on VA ECMO: decrease blood flow while maintaining some sweep gas flow. (Modified from [Ody_Stocker/Shutterstock.com](https://www.shutterstock.com/user/ody-stocker).)

After weaning down blood flow to a minimal level (say 1.5 L/min to 2.5 L/min), the ECMO circuit is sometimes clamped to examine the effect of cessation of blood flow. This can only be done for a brief period as static blood can increase the risk of thrombosis and embolization.

Sweep gas flow should be weaned down to a minimal level prior to weaning off ECMO. The maintenance of some sweep gas flow is absolutely necessary, and titrating off sweep gas flow

completely will deoxygenate any blood flow that is going through the ECMO circuit. Even though it was already mentioned, it is an essential point, so indulge me.

Never wean off sweep gas flow completely on VA ECMO. Doing so will lead to rapid decompensation.

Ok, now I'm satisfied.

WEANING IN VV ECMO

As opposed to VA ECMO, perfusion to end organs on VV ECMO is maintained completely by the native cardiac output. Therefore, the effect of weaning ECMO blood flow is less impactful in informing whether the patient is ready to be weaned. Blood flow may be able to be titrated up or down throughout the course as mentioned in the previous chapter, but the true test of whether the patient is ready to come off VV ECMO is ultimately the ability to wean off sweep gas flow (Fig. 18.5).

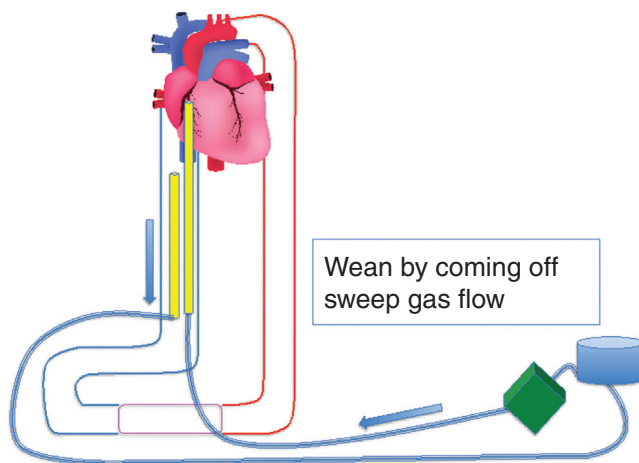


FIG. 18.5 Weaning on VV ECMO: come off sweep gas flow completely while maintaining ECMO blood flow. (Modified from [Ody_Stocker/Shutterstock.com](https://www.shutterstock.com/user/ody_stocker).)

The advantage of maintaining ECMO blood flow while trialing off sweep gas involves getting a sense of what the patient is likely to look like off ECMO since any oxygenation/ CO_2 clearance will be completely due to the performance of the native lungs. Additionally, since the blood flow through the ECMO circuit is maintained, longer trial periods can be maintained to examine the effect of weaning over the course of hours if needed.

The first thing to look for while weaning is oxygen saturation. While the patient may tolerate minimal sweep and appears ready to be weaned off ECMO, it is important to realize that the oxygenation of blood is maintained as long as any sweep gas is present. Turning off the sweep gas may lead to a drastic change if the patient is not ready to come off ECMO. This usually manifests as an acute drop in oxygen saturation within the first couple of minutes after sweep gas flow is turned off.

If oxygen saturation is maintained, the next variable to assess is work of breathing. A patient may be able to maintain oxygenation but may be tachypneic/dyspneic/fatigued in order to do so. This clinical constellation often means that the patient is improving but not yet ready to wean.

If a patient is able to maintain adequate oxygen saturation with a normal work of breathing, the final variable to monitor is CO_2 level. The ability to clear CO_2 sometimes lags behind the ability to oxygenate; therefore elevated PaCO_2 levels on ABG that cannot be optimized with lung-protective strategies may indicate a failure to wean.

The decision to wean must be weighed against the risks and benefits of staying on ECMO. Patients with complications related to ECMO (bleeding, infection, cannula issues) may warrant more aggressive weaning from ECMO.

That said, one final note about weaning on VV ECMO. Due to longer ECMO run times, the temptation may exist to push weaning in order to get a patient off ECMO. If not limited by complications, weaning often happens when the patient is ready to come off. Patience is key.

WEANING FiO_2 TO SWEEP GAS FLOW

We mentioned that FiO_2 is the final variable that can be manipulated with regards to sweep gas flow. Often this is set at 100% and left at 100% until the patient is able to be weaned off entirely.

FiO_2 can be weaned when PaO_2 of the arterial blood gas is high ($\text{PaO}_2 > 120$ mmHg), as hyperoxia can potentially be toxic. However, for the most part, while on ECMO support, PaO_2 is either low or normal. Weaning FiO_2 to sweep gas flow should be avoided if this means increasing the amount of oxygen administered to the native lungs, which often comes at a greater cost of toxicity.

The presence of high PaO_2 on the arterial blood gas is more likely to occur on VA ECMO, particularly when central cannulated. Adjustments in FiO_2 to sweep gas flow are more common in this mode of support.

Rarely, FiO_2 can be weaned as a last phase of weaning on VV ECMO, particularly in patients who are on minimal sweep gas flow, but who do not tolerate the final step off of trial completely off sweep. FiO_2 weaning in these patients may allow for a more gradual weaning in this select group of patients.

PUTTING IT TOGETHER

Some aspects of ECMO care are relatively consistent across different teams and programs, while other aspects involve practices that vary widely. The interpretation of CO_2 clearance, adjustment of sweep gas flow, and practices for weaning ECMO definitely fall into the latter category. Different teams and providers may employ different techniques for adjustment and titration of sweep gas flow – some making large changes with frequent adjustments while others employ a very conservative approach.

Ultimately, the right technique is that which leads to a good clinical response and ultimately a good outcome to the patient. Whenever possible, assess the clinical effect of changes in sweep gas flow. This will not only allow you to hone your skills of sweep gas flow titration but also contextualize your changes to a meaningful outcome for the patient under your care.

SUGGESTED READING

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